STATISTICS 801

Fall 2016

Homework Assignment 4
Due Oct 19, 8:00 am

1. Lead poisoning results from an unnatural craving for such substances as paint and affects a large number of children each year causing them to suffer severe retardation. But it has also been investigated whether such cravings have a nutritional explanation. In a study, each rat in a group of 20 were randomly assigned to a normal diet (control group) or to a calcium-deficient diet(experimental group). Each rat occupied a separate cage and was monitored to observe the quantity of a .15% lead acetate solution consumed. The data are:

i	Group	y_i, Quantity of .15% $Pb(C_2H_3O_2)_2$ consumed										n	$\sum y$	$\sum y^2$
1	Control	5.4	6.2	3.1	3.8	6.5	5.8	6.4	4.5	4.9	4.0	10	50.6	268.76
2	Experimental	8.8	9.5	10.6	9.6	7.5	6.9	7.4	6.5	10.5	8.3	10	85.6	752.22

`Note:` Use hand calculation for parts (a) to (c) and show work. Use an appropriate software analysis (use your favorite statistical software) of the data for answering parts (d) to (f). `Turn in the marked output with your comments` and show work.

(a) Calculate the sample means $\bar{y}_1$, $\bar{y}_2$, the sample variances s_1^2, s_2^2, and the pooled standard deviation s_p by hand, using a calculator and the summaries given above.

(b) Do the data provide sufficient evidence that that the mean quantity of lead acetate consumed by the rats in the experimental group is greater than that consumed by the rats in the control group? First, state H_0 and H_a in terms of the difference in the mean lead acetate consumed in the two populations, μ_1 and μ_2, respectively. Use a two-sample t-test assuming equal variances for the two populations, and make a decision using $\alpha = .05$.

(c) Calculate a 90% confidence interval for the difference in population means in lead consumption between control and experimental groups. Assume equal variance for the two populations. Use this interval to test the hypothesis you stated in part (b). State the α-level of this test and your conclusion.

(d) Assess the validity of the three conditions necessary for the test used in part (b) to be valid. Give reasons for assuming that these are satisfied or not to an acceptable degree?

(e) On your output, circle the t-statistic for testing the hypothesis you stated in Part (b)and the p-value associated with it in your output. Extract them from your output and write them here also.

(f) Extract the 90% confidence interval for the difference in population means as in Part (c) from your output. Write it here and circle it in the output.

2. How quickly do synthetic fabrics such as polyester decay in landfills? A researcher in environmental studies buried polyester strips in the soil for different lengths of time, then dug up the strips and measured the force required to to break them. Breaking strength is an easy measure and is a good indicator of decay. Lower strengths mean the fabric has decayed more. Part of the study involved burying 16 polyester strips in well-drained soil in the summer at the same depth at random locations. Eight of the strips were chosen at random to be dug up after 2 weeks and the other 8 were dug up after 16 weeks. The breaking strengths were measured in pounds of each strip.

i	Time	y_i, Breaking Strength, lbs.								n	$\sum y$	$\sum y^2$
1	2 weeks	118	135	127	120	129	123	115	130	8	997	124,573
2	16 weeks	125	98	113	136	110	121	118	105	8	926	108,184

`Note:` Use hand calculation for parts (a) to (c) and show work. Use an appropriate software analysis of the data for answering parts (d),(e), and (f). `Turn in the marked output` and show work.

(a) Calculate the sample means $\bar{y}_1$ and $\bar{y}_2$ and sample variances s_1^2 and s_2^2, respectively, of the two samples. Draw a back-to-back stem-and-leaf plots of the data. Is there evidence to believe that the *variances* of the breaking strengths for the two populations of polyester strips may be different? Explain using only the sample statistics and the plot.

(b) Calculate, by hand, Welch's two-sample t-statistic and perform a hypothesis test to find out whether the observed difference in the sample means is good evidence that polyester decays more in 16 weeks than in 2 weeks. Use $\alpha = 0.05$. State H_0 and H_a in terms of appropriate `population parameters you define and label clearly`. Interpret your findings and show your work.

(c) Compute a 90% confidence interval for the difference in population means you defined in part (b) for the two populations assuming unequal population variances.

(d) Report evidence you find from your output (a) for supporting the normality assumption of the two samples, and (b) for supporting the conclusion that the population variances are not equal.

(e) The t-statistic for testing the hypothesis you stated in Part (b) under the unequal variances assumption can be computed in a statistical software (for example SAS or R). Circle it in your output and copy here the statistic and the p-value associated with it extracted from the output.

(f) Extract the 90% confidence interval for the difference in population means as in Part (c) computed under unequal variances assumption from the output. Copy it here and circle it in the output.

3. A local doctor suspects that there is a seasonal trend in the occurrence of the common cold. He estimates that 40% of the cases each year occur in the winter, 40% in the spring, 10% in the summer, and 10% in the fall. The doctor collected the following information from a random sample of 1,000 cases of patients with the common cold over the past year.

Season	Frequency
Winter	374
Spring	292
Summer	169
Fall	165

Would you agree with the doctor's estimates, based on the sample information? State the null and alternative hypotheses. Perform a statistical test using $\alpha = 0.05$. Draw conclusions.

4. In random sampling, and interviewer asked 40 newspaper editors their opinions on the degree of future suppression of freedom of the press brought about by recent court decisions. The editor's opinions are summarized in the table below.

Degree of Suppression	Frequency
None	8
Very little	8
Moderate	10
Severe	14

Use these data to test the null hypothesis that each category is equally preferred. Use $\alpha = 0.05$. Draw conclusions from these data. What reservation(s), if any, do you have concerning your conclusions?

5. The following observations were fracture toughness (in $ksi\sqrt{in}$) measured on 22 base plates made of 18% nickel maraging steel: 79.7, 79.9, 93.7, 83.7, 82.2, 75.8, 79.6, 69.5, 75.7, 80.1, 76.1, 71.9, 75.5, 78.1, 72.6, 76.2, 77.9, 73.5, 73.1, 76.2, 77.0, 73.3. Use this data to provide answers to the following questions.

   `Note:` Do all computations for parts (b), and (c), using hand calculation. For this data $\sum y = 1701.3$, $\sum y^2 = 132,097.35$. Use your output to provide answers to part (a). `Turn in the marked output and show work.`

   (a) Do the box plot or the normal probability plot indicate any violation of the conditions underlying the use of the chi-square procedures for constructing confidence intervals or testing hypotheses about σ?

   (b) A quality control inspector might require the fracture toughness of the steel plates to be reasonably consistent, in particular, that σ of more than $5ksi\sqrt{in}$ to be unsatisfactory. Is there sufficient evidence in the data to determine that the standard deviation meets this criterion? State appropriate hypotheses and perform a test with $\alpha = .05$.

   (c) Estimate the standard deviation of the fracture toughness of the steel plates using a 95% confidence interval.

6. A consumer protection magazine was interested in comparing tires purchased from two different companies that each claimed their tires would last 40,000 miles. A random sample of 10 tires of each brand was obtained and tested under simulated road conditions. The number of miles until the tread thickness reached a specified depth was recorded for all tires. The data are given (in 1,000 miles) below:

Brand	*Miles(in thousands)*										$\sum y$	$\sum y^2$
I	38.9	39.7	42.3	39.5	39.6	35.6	36.0	39.2	37.6	39.5	387.9	15081.01
II	44.6	46.9	48.7	41.5	37.5	33.1	43.4	36.5	32.5	42.0	406.7	16820.63

   (a) Use the normal probability plots output from your output to comment on whether the assumption of normality of the two populations is reasonable. Does this plot support the assumption that the population variances are the same for the two populations? Explain

   (b) Compute the sample variances s_{I}^2 and s_{II}^2, respectively, of the two samples using the above summary statistics. (`Must do hand computation`)

   (c) Compute a statistic to test the research hypothesis that the two population variances of wear rates for the two brands are different using $\alpha = .05$ by hand. State your hypotheses, the rejection region and your conclusion. (`Must do hand computation`)